

Notes on Category Theory

from Hilton and Wu, A Course in Modern Algebra

Notations

$\mathfrak{C}, \mathfrak{D}$: categories

F, G : functors

$\text{Ob}(\mathfrak{C})$: the set of objects of category $\mathfrak{C}$

$\text{Mor}(\mathfrak{C})$: the set of morphisms of category $\mathfrak{C}$

$\mathfrak{C}(X, Y)$: the set of morphisms from object X to object Y

$\mathfrak{S}$: the category of sets

$\mathfrak{G}$: the category of groups

$\mathfrak{T}$: the category of topological spaces and continuous functions

$\mathfrak{Ab}$: the category of abelian groups

$\mathfrak{D}_K$: the category of finite dimensional vector spaces over the field K

Categories

Axioms of Categories:

CAT-1 The sets $\mathfrak{C}(A, B)$ and $\mathfrak{C}(C, D)$ are disjoint unless $(A, B) = (C, D)$.

CAT-2 $f : A \rightarrow B, g : B \rightarrow C, h : C \rightarrow D$, then $h(gf) = (hg)f$

CAT-3 For any object A , there is an identity morphism $1_A : A \rightarrow A$ such that for any other two morphisms $f : A \rightarrow B, g : C \rightarrow A, f1_A = f, 1_Ag = g$.

Definition 1 (Initial and Terminal Objects): An object I is said to be *initial*, if the set of morphisms $\mathfrak{C}(I, -)$ is a singleton. An object T is said to be *terminal (coinitial)* if the set of morphisms $\mathfrak{C}(-, T)$ is a singleton, where $-$ is any object in $\mathfrak{C}$. If an object is both initial and terminal it is called a zero.

Example: $\{\}$ is the initial object in $\mathfrak{S}$. 1 is the zero object in $\mathfrak{G}$.

Remark: A zero object is also called *universal*.

Functors

Definition 2 (Forgetful functor): A functor $U : \mathfrak{C} \rightarrow \mathfrak{S}$ where $\mathfrak{S}$ is the category of the underlying set of $\mathfrak{C}$ and $U(f)$ is a function in $\mathfrak{S}$ such that $f(X) = U(f)(U(X))$. For example, $U : \mathfrak{D}_K \rightarrow \mathfrak{Ab}$ is a forgetful functor "forgetting" the multiplication operation in $\mathfrak{D}_K$.

Definition 3 (Faithful and Full Functors): A functor $F : \mathfrak{C} \rightarrow \mathfrak{D}$ induces a map $F^* : \mathfrak{C}(X, Y) \rightarrow \mathfrak{D}(FX, FY)$. A faithful functor F (resp. full functor F) induces an injective (resp. surjective) map F^* for all X, Y .

Example: The forgetful functor $U : \mathfrak{G} \rightarrow \mathfrak{S}$ is faithful because all homomorphisms are functions between the sets underlying a group and the map $\varphi \mapsto U(\varphi)$ is injective. However, it is not full.

Natural Transformations

Definition 4 (Natural Transformations): A map $\tau : F \rightarrow G$ is called a natural transformation if it assigns to each object $X \in \mathfrak{C}$ a morphism τ_X such that the following diagram commutes:

$$\begin{array}{ccc}
FX & \xrightarrow{\tau_X} & GX \\
F(f) \downarrow & & \downarrow G(f) \\
FY & \xrightarrow{\tau_Y} & GY
\end{array}$$

Example: In $\mathfrak{D}_K$, we have

$$\begin{array}{ccc}
V & \xrightarrow{v \mapsto \tilde{v}} & V^{**} \\
f \downarrow & & \downarrow f^{**} \\
W & \xrightarrow{w \mapsto \tilde{w}} & W^{**}
\end{array}$$

where $v \mapsto \tilde{v}$ is the isomorphism between V and the double dual space V^{**} and $(f^{**}(v^{**}))(\phi) = v^{**}(f^*(\phi))$ and f^* is the dual map of f and $\phi \in W^*$. Here $\tau : \text{id} \rightarrow (**)$.

Duality Principle

We say the category $\mathfrak{C}^{\text{op}}$ the *opposite category* of $\mathfrak{C}$ if $\text{Ob}(\mathfrak{C}) = \text{Ob}(\mathfrak{C}^{\text{op}})$ and $\mathfrak{C}(X, Y) = \mathfrak{C}^{\text{op}}(Y, X)$ such that $f \circ^{\text{op}} g = g \circ f$.

An example of dual statement S : $S(\mathfrak{C})$: I is an initial object if $\mathfrak{C}(I, X)$ is a singleton for every $X \in \mathfrak{C}$.
 $S(\mathfrak{C}^{\text{op}})$: I is an initial object if $\mathfrak{C}^{\text{op}}(I, X)$ is a singleton for every $X \in \mathfrak{C}^{\text{op}}$
 $S^{\text{op}}(\mathfrak{C})$: I is a coinital object if $\mathfrak{C}(X, I)$ is a singleton for every $X \in \mathfrak{C}^{\text{op}}$.

If S is a statement such that it is meaningful in all categories. If S is true in $\mathfrak{C}$, then S is also true in $\mathfrak{C}^{\text{op}}$

Definition 5 (Monic and Epic): A morphism f is called *monic* if $f \circ u = f \circ v$ implies $u = v$. We call the dual analoge *epic*.

Theorem 1:

$$\begin{aligned}
\text{monic} &\Leftrightarrow \text{injective} \\
\text{epic} &\Leftrightarrow \text{surjective}
\end{aligned}$$

Remark: The cokernel $\text{Coker } f = Y / \text{im } f$ and $\ker f$ are dual pairs. Hence, if f is surjective in $\mathfrak{C}$, then f is injective in $\mathfrak{C}^{\text{op}}$.

Definition 6 (Kernel), In category theory, a kernel κ of a morphism $f : X \rightarrow Y$ of a category $\mathfrak{C}$ is defined as a monic morphism $\kappa : K \rightarrow X$ such that

- (i) $f \circ \kappa = 0$
- (ii) $f \circ g = 0$ implies the existence of h such that $g = f \circ h$.

Definition 7 (Contravariant Functor): A functor $F : \mathfrak{C} \rightarrow \mathfrak{D}$ is called contravariant if $F : \mathfrak{C}^{\text{op}} \rightarrow \mathfrak{D}$.

Example (*The contravariant powerset functor*): Let $Q : \mathfrak{S} \rightarrow \mathfrak{S}$ such that $QX = 2^X$ and for a morphism $f : X \rightarrow Y$, $(Qf)B = f^{-1}B$ where $B \subseteq Y$ so that $Qf : 2^Y \rightarrow 2^X$. This gives rise to

more interesting properties than the *covariant powerset functor* P . For example, consider the subsets, $A_1, A_2 \subseteq X$ and $B_1, B_2 \subseteq Y$. It is not necessarily true that

$$P(f)(A_1 \cap A_2) \stackrel{?}{=} P(A_1) \cap P(A_2)$$

where as, the following is

$$Q(f)(B_1 \cap B_2) = Q(B_1) \cap Q(B_2)$$

for example if $A_1 = \{x\}, A_2 = \{y\}, f(x) = f(y)$.

Products and Coproducts

Definition 8 (Product): If $A_1, A_2 \in \text{Ob}(\mathfrak{C})$, we define (P, p_1, p_2) with $p_i : P \rightarrow A_i$ if for any $f_i : X \rightarrow A_i$ there is a unique $f : X \rightarrow P$ following diagram commutes:

$$\begin{array}{ccc} & & A_1 \\ & \nearrow f_1 & \uparrow p_1 \\ X & \xrightarrow{f} & P \\ & \searrow f_2 & \downarrow p_2 \\ & & A_2 \end{array}$$

Products in categories are unique upto canonical equivalence. Example: $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \in \mathfrak{C}$ does not exist in the category of cyclic groups.

Proposition 1:

- (i) $(g_1 \times g_2) \circ (h_1 \times h_2) = (g_1 \circ h_1 \times g_2 \circ h_2)$.
- (ii) $(g_1 \times g_2) \circ \{f_1, f_2\} = \{g_1 \circ f_1, g_2 \circ f_2\}$
- (iii) $\{f_1, f_2\} \circ g = \{f_1 g, f_2 g\}$

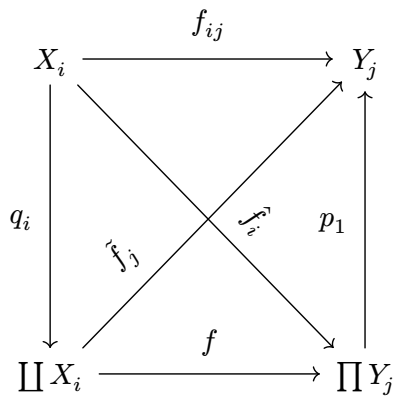
Example: In the category of finitely generated abelian groups, no infinite set has a product. The set $\{\mathbb{Z}/p^n\mathbb{Z} : n \in \mathbb{N}\}$ has infite product in the category of torision abelian groups namely the torsion subgroup of the product $\prod \mathbb{Z}/p^n\mathbb{Z}$.

Definition 9 (Coproducts): (Q, q_1, q_2) such that the below diagram commutes.

$$\begin{array}{ccc} & & A_1 \\ & \nwarrow f_1 & \downarrow q_1 \\ X & \xleftarrow{f} & Q \\ & \swarrow f_2 & \uparrow q_2 \\ & & A_2 \end{array}$$

One usually denotes the coproduct of A and B by $A \coprod B$.

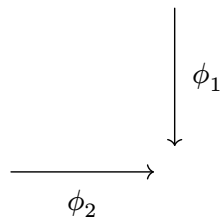
Theorem 2: Let $X_1, X_2, \dots, X_m, Y_1, \dots, Y_n \in \text{Ob}(\mathfrak{C})$. Let $f_{ij} : X_i \rightarrow Y_j \in \mathfrak{C}(X_i, Y_j)$ and consider the below diagram:



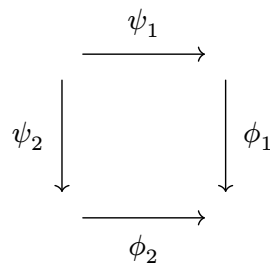
Where $\hat{f}_i$ is the unique morphism from X_i associated with the product $\prod Y_i$ and $\tilde{f}_j$ is the unique morphism to Y_j associated with the coproduct $\coprod X_i$. $f = \hat{f} = \tilde{f}$.

Pullback and Pushouts

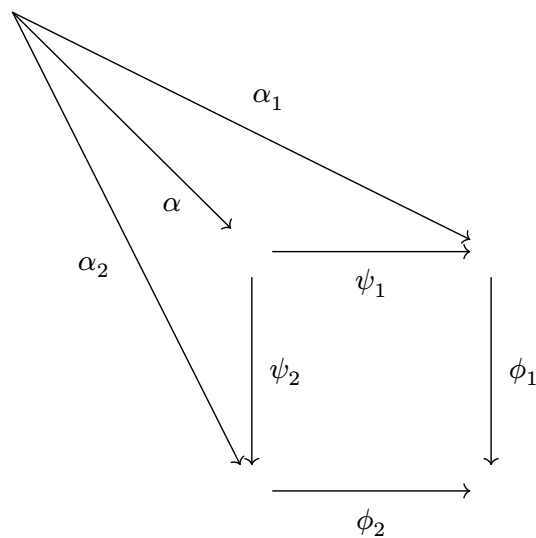
Definition 10 (Pullback and Pushout): A pullback of the diagram



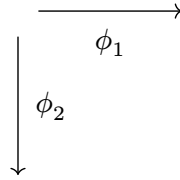
is the commutative square



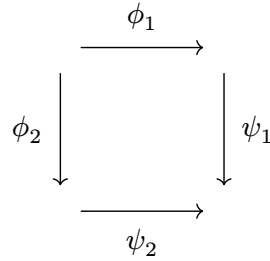
such that the following universal property holds:



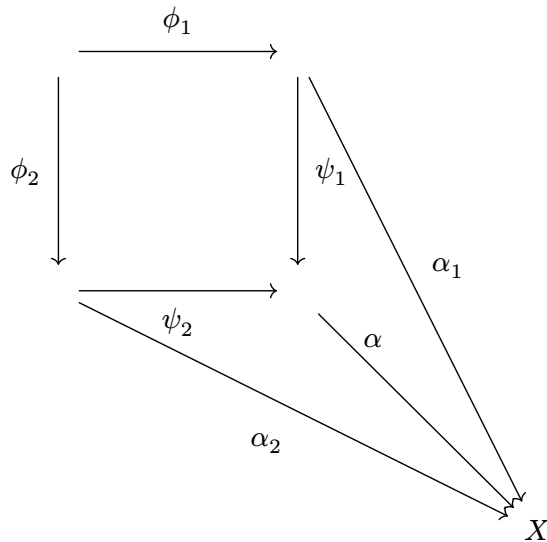
A pushback of the diagram



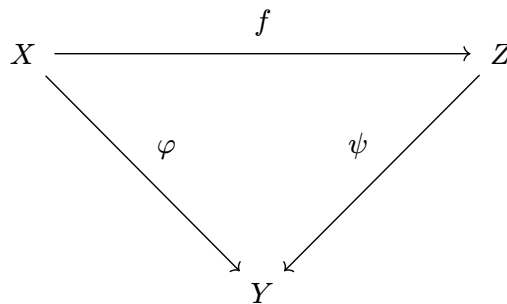
is the commutative square



such that the following universal property holds:



The pullback is similar to the product in the following way. Consider a category $\mathfrak{C}$. Let Y be an object in $\mathfrak{C}$. Then define the category $\mathfrak{C}/Y$ as the category of morphisms $f : X \rightarrow Y$ for all $X \in \text{Ob}(\mathfrak{C})$. The morphisms of $\mathfrak{C}/Y$, $f : \varphi \rightarrow \psi$ are the morphisms such that the following diagram commutes:



Then the pullback of φ and ψ in $\mathfrak{C}$ is the product of φ and ψ in the category $\mathfrak{C}/Y$.

Theorem 3: Suppose $\mathfrak{C}$ is a category with zero and consider the following pullback diagram of the $\phi_1 : X_1 \rightarrow Y$ and $\phi_2 : X_2 \rightarrow Y$:

$$\begin{array}{ccc}
X_0 & \xrightarrow{\psi_1} & X_1 \\
\psi_2 \downarrow & & \downarrow \phi_1 \\
X_2 & \xrightarrow{\phi_2} & Y
\end{array}$$

then the following statements are true:

- (i) If κ is a kernel of ψ_2 , then ϕ_1 has a kernel $\psi_1\kappa$.
- (ii) If λ is a kernel of ϕ_1 , then $\lambda = \psi_1\kappa$ for some kernel κ of ψ_2 .

Sketch of proof: One can use the pullback property that if $\phi_1\alpha = \phi_2\beta$, then there is a unique morphism $\kappa : - \rightarrow X_0$, that makes $\psi_1\kappa$ a kernel of ϕ_1 and κ a kernel of ψ_2 .

Proposition 2: In $\mathfrak{Ab}$, the diagram

$$\begin{array}{ccc}
A & \xrightarrow{\psi_1} & X_1 \\
\psi_2 \downarrow & & \downarrow \phi_1 \\
X_2 & \xrightarrow{\phi_2} & Y
\end{array}$$

is pullback if and only if $\hat{\psi}$ is the kernel of $\tilde{\phi}$ in $A \xrightarrow{\hat{\psi}} X_1 \oplus X_2 \xrightarrow{\tilde{\phi}} Y$

Adjoint Functors

For $B \in \text{Ob}(\mathfrak{S})$, let $F = - \times B$ and $G = -^B$ be two functors from $\mathfrak{S} \rightarrow \mathfrak{S}$. We then have the equivalence of the following sets,

$$\mathfrak{S}(FA, C) \cong \mathfrak{S}(A, GC)$$

If $\alpha : A_1 \rightarrow A$ and $\beta : C \rightarrow C_1$ are two morphisms in $\mathfrak{S}$, and $\phi : FA \rightarrow C$, then we have the equivalence η ,

$$\begin{array}{ccccccc}
FA_1 & \xrightarrow{F\alpha} & FA & \xrightarrow{\phi} & C & \xrightarrow{\beta} & C_1 \\
& & \downarrow \eta & & & & \\
A_1 & \xrightarrow{\alpha} & A & \xrightarrow{\eta(\phi)} & GC & \xrightarrow{G\beta} & GC_1
\end{array}$$

which is natural as in

$$\eta(\beta \circ \phi \circ F\alpha) = G\beta \circ \eta(\phi) \circ \alpha.$$

To show this (informally), it suffices to prove $\eta(\beta \circ \phi) = G\beta \circ \eta(\phi)$ and $\eta(\phi \circ F\alpha) = \eta(\phi) \circ \alpha$. Since $\eta((a, b) \mapsto c) = a \mapsto (b \mapsto c)$, $\eta(\beta \circ \phi) = \eta\left((a, b) \mapsto c \xrightarrow{\beta} c'\right) = a \mapsto b \mapsto c \xrightarrow{\beta} c'$ and the first equation follows. Similarly, $\eta(\phi \circ F\alpha) = \eta\left((a', b) \xrightarrow{\alpha} (a, b) \mapsto c\right) = a' \xrightarrow{\alpha} a \mapsto b \mapsto c$.

Definition 11 (Adjoint Functor): A functor $F : \mathfrak{C} \rightarrow \mathfrak{D}$ is called *left adjoint* to functor $G : \mathfrak{D} \rightarrow \mathfrak{C}$ if there exists a natural equivalence η , such that

$$\mathfrak{D}(FA, B) \cong^{\eta} \mathfrak{C}(A, GB).$$

We also say G is right adjoint to F and we say adjugant η establishes an adjunction. We write $F \stackrel{\eta}{\vdash} G$.

Example: $\text{Abel} \vdash U$, i.e.,

$$\mathfrak{Ab}(G/G', A) \cong \mathfrak{G}(G, UA)$$

In fact the morphism $\phi \mapsto \eta(\phi) = U\phi$ is an isomorphism.

Theorem 4: If $F_1 \vdash G_1$ and $F_2 \vdash G_2$, then $F_1 F_2 \vdash G_2 G_1$.

Sketch of proof: Since $\mathfrak{D}(F_1 A, B) \cong \mathfrak{C}(A, G_1 B)$ and $\mathfrak{C}(F_2 A, B) \cong \mathfrak{E}(A, G_2 B)$, $\mathfrak{D}(F_1 F_2 A, B) = \mathfrak{C}(F_2 A, G_1 B) = \mathfrak{E}(A, G_2 G_1 B)$

Theorem 5: A functor determines its adjoint upto natural canonical equivalence.

Theorem 6: If a functor has a left adjoint, it preserves, products, kernels, pullbacks and monics.

Definition 12 (units and counits): Let $F \vdash G$, then the unit of the adjunction is the natural transformation $\varepsilon : 1 \rightarrow GF$ and the counit of the adjunction is the natural transformation $\delta : FG \rightarrow 1$. We have $\varepsilon_A = \eta(1_{FA})$ and $\delta_B = \eta^{-1}(1_{GB})$.

Theorem 7: If $F : \mathfrak{C} \rightarrow \mathfrak{D}$ and $G : \mathfrak{D} \rightarrow \mathfrak{C}$ are two functors and $\varepsilon : 1 \rightarrow GF$ and $\delta : FG \rightarrow 1$ are two natural transformations, such that

$$\begin{array}{ccccc} F & \xrightarrow{F\varepsilon} & FGF & \xrightarrow{\delta F} & F \\ \downarrow & & \downarrow & & \uparrow \\ & & 1 & & \\ \\ G & \xrightarrow{\varepsilon G} & GFG & \xrightarrow{G\delta} & G \\ \downarrow & & \downarrow & & \uparrow \\ & & 1 & & \end{array}$$

The rule $\eta(f) = Gf \circ \varepsilon_A$ where $f : FA \rightarrow B$ defines an adjunction $F \stackrel{\eta}{\vdash} G$.